

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November 2013

CL 202.01 Strength of Materials

Date: 23.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the followings:

1. What is section modulus if a rectangular section having width (b) and depth (d)? [01]
2. In a triangular section, the maximum shear stress occurs at _____. [01]
(a) Apex of the triangle (b) mid of the height (c) 1/3 of the height
3. When a rectangular section of a beam is subjected to a shearing force, the ratio of maximum shear stress to the average shear stress is _____. [01]
4. What is section modulus of a circular section of diameter (D)? [01]
5. Draw qualitative shear stress distribution diagram of symmetrical I- Section. [01]
6. A wooden bar is subjected to a tensile stress of 5 MPa. What will be the values of normal and shear stresses across a section, which makes an angle of 25° with the direction of the tensile stress? [02]

- Q - 2 (a)** A cantilever timber beam of 500 mm length carries a UDL of 5 kN/m from support to mid-span. It also carries a point load of 6 kN at the tip of the cantilever. If the beam section is 50 mm wide and 150 mm deep throughout the length, determine maximum bending stress. Neglect the self weight of the beam. [04]
- (b)** Determine the dimensions of joist of a timber for span 8 m to carry a brick wall 200 mm thick and 5 m high, if the density of brick work is 1850 kg/m^3 and the maximum permissible stress is limited to 7.5 MN/m^2 . Given that the depth of joist is twice the width. [04]
- (c)** A beam in the form of cross is shown in **Figure 1** is subjected to shear force of 100 kN. Sketch the shear stress distribution diagram. [06]

OR

- Q - 2 (a) A beam of symmetrical section, depth = 400 mm, $I = 19300 \text{ cm}^4$ is simply supported over a span of 8 meters. What uniformly distributed load may it carry if the maximum bending stress is not to exceed 120 N/mm^2 ? What concentrated load may be carried by the beam at the centre with the same permissible bending stress? [03]
- (b) A flitched beam consists of two wooden joist 150 mm wide and 300 mm deep and a steel plate 10 mm thick and 250 mm deep placed symmetrically between wood sections. If the maximum stress in the wooden joist is 10 N/mm^2 . Find the corresponding maximum stress attained in steel. Find also the moment of resistance of the section. Take $E_s = 20 E_w$. [05]
- (c) A beam in the form of cross is shown in Figure 2 is subjected to shear force of 25 kN. Draw the shear stress distribution diagram. [06]

Q - 3 Attempt any Two:

[14]

- (a) A small block 40 mm long $\times$ 30 mm high and 5 mm thick is subjected to uniformly distributed tensile forces having the resultant values shown in Figure 3. Compute the stress component developed along the diagonal AC.
- (b) At a certain point in a strained material, normal stresses of 128 MN/m^2 , tension, and 100 MN/m^2 , compression are acting on planes right angles to each other. The greater principal stress in the material is limited to 160 MN/m^2 . To what shearing stress may the material be subjected on the given planes, and what will be the maximum shearing stress at that point?
- (c) At a certain point in a strained material, the values of normal stresses across two planes at right angles to each other are 80 MPa and 32 MPa, both tensile and there is a shear stress of 32 MPa clockwise on the plane carrying 80 MPa stress across the planes as shown in Figure 4.
- (i) Locate the principal planes and evaluate the principal stresses
- (ii) Calculate the maximum shear stress and specify its plane

SECTION - II

- Q - 4 (a) Define point of contra flexure. [01]
- (b) A solid shaft transmits a torque of 50 kNm. If the angle of twist is limited to 1.5° in 2 m length of shaft and the shear stress to 80 N/mm^2 , determine the diameter of shaft. Take $G = 0.8 \times 10^5 \text{ N/mm}^2$ [03]
- (c) An isosceles triangular section ABC has base width 80 mm and height 60 mm. [03]

Determine the moment of inertia of the section about the center of gravity of the section and the base BC.

- Q - 5 (a) Determine the moment of inertia of the shaded area of a semi-circle as shown in **Figure 5** about its base AB. [04]
- (b) Compute the moment of inertia of the cross-section as shown in **Figure 6** about axis AA. [05]
- (c) A composite shaft is made up of brass rod of 40 mm diameter enclosed in a steel tube of external diameter 50 mm and 5 mm thick. The shafts are rigidly fixed to each other so that there is no relative motion between the two. The shaft is required to transmit a torque of 4 kNm. Determine the stresses developed in brass and steel if both shafts have equal lengths. Take $G_{\text{brass}} = 40 \text{ GPa}$ and $G_{\text{steel}} = 80 \text{ GPa}$. [05]

OR

- Q - 5 (a) A rectangular hole is made in a triangular section as shown in **Figure 7**. Determine the moment of inertia of the section about base BC. [04]
- (b) **Figure 8** shows the cross-section of a cast iron beam. Determine the moment of inertia of the section about vertical axes (I_{yy}) passing through the centroid of the section. [05]
- (c) A 60 mm diameter shaft transmits 80 kW at 100 rpm. The shaft is connected to a machine component by means of key which is 20 mm wide and 100 mm long. Find the shear stresses developed in the shafts and the key. [05]

Q - 6 Attempt any Two:

[14]

- (a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 9**.
- (b) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 10**.
- (c) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 11**.

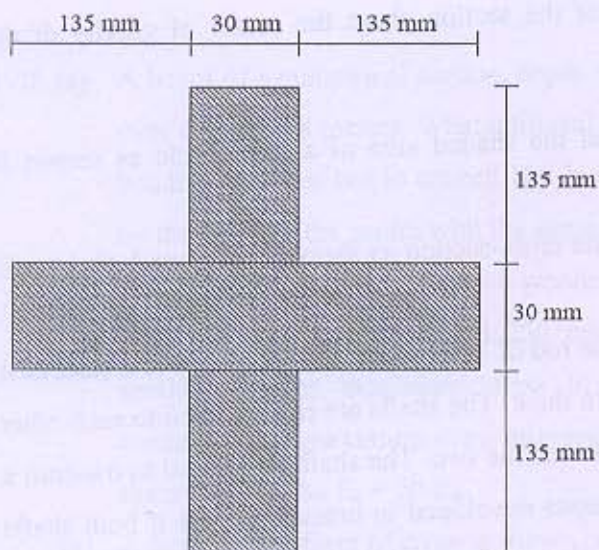


Figure 1

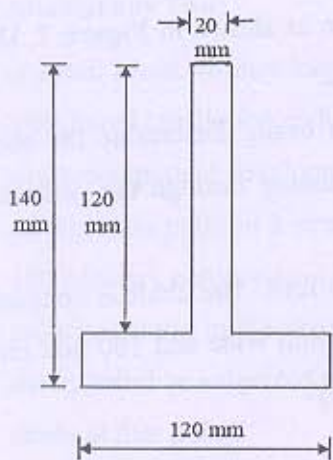


Figure 2

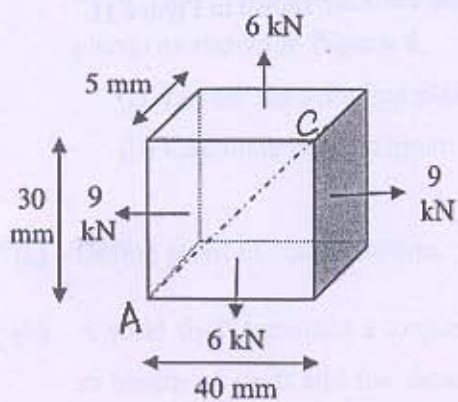


Figure 3

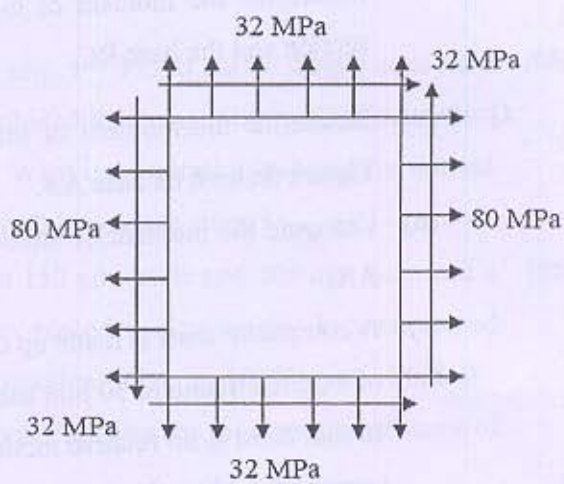


Figure 4

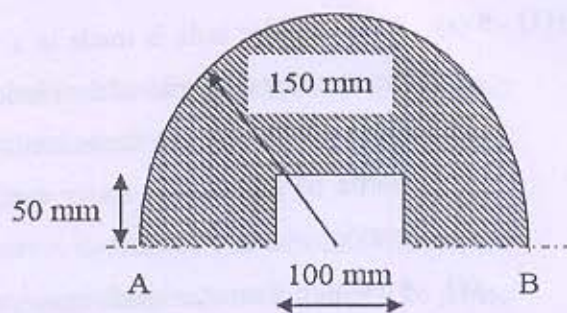


Figure 5

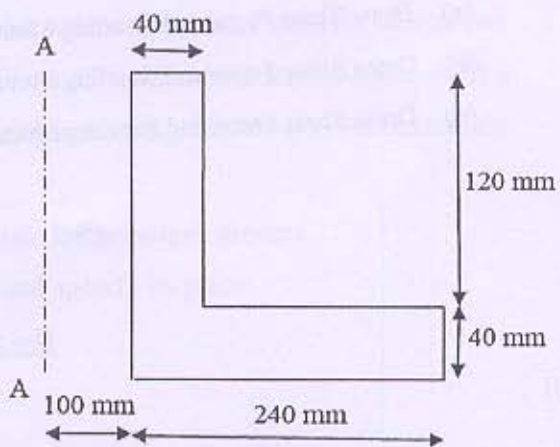


Figure 6

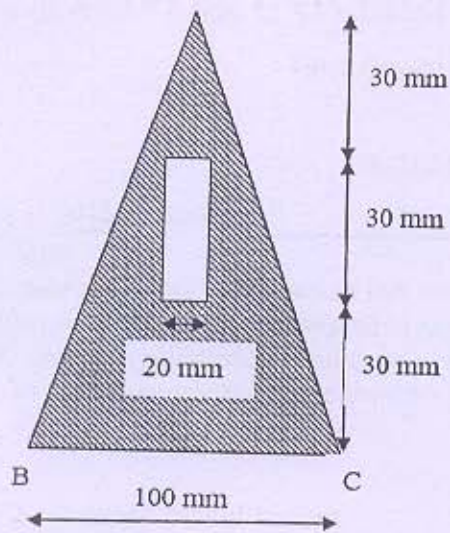


Figure 7

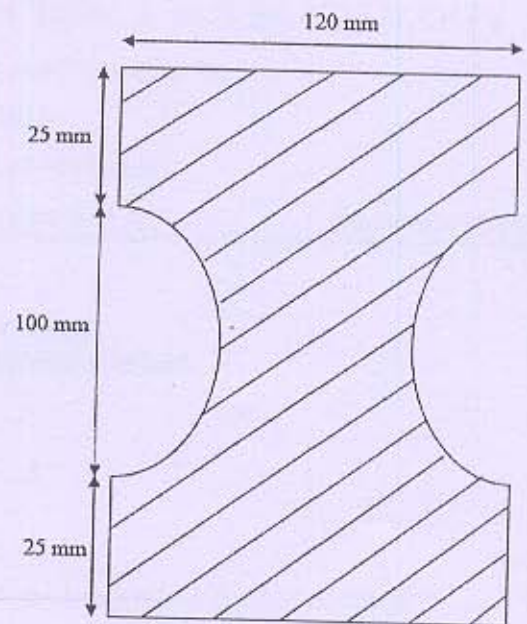


Figure 8

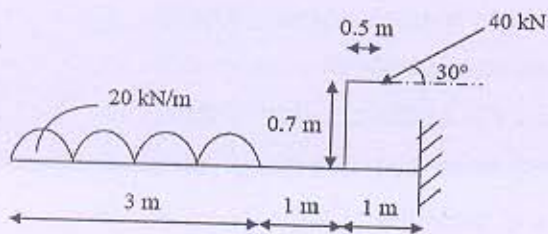


Figure 9

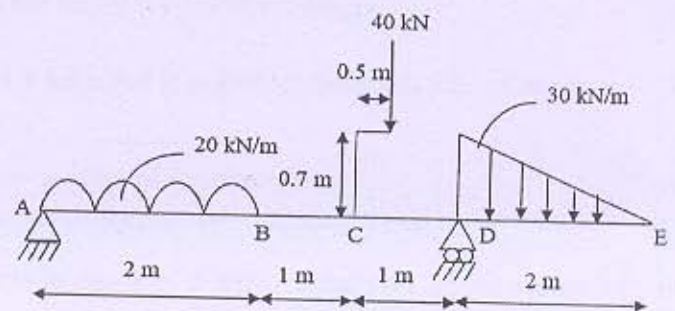


Figure 10

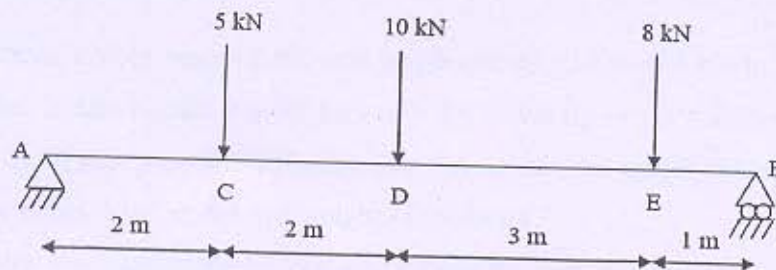


Figure 11